

**P27 -Development of an AI-Assisted Collaborative Worldbuilding Platform
for Game Design**

Group Formal Tag: COMP3888_M17_02_ZL

Tutor Name: Zhenlin Liu

Client Name: Frank Lai

Group members details

No.	Student Name	Student ID	Student Unikey	XP-Roles
1	Shuhao Pei	530748773	spei0511	Manager
2	Jiayue Wu	530816072	jiwu0646	Doomsayer
3	Haoyu Deng	540727454	hden0618	Customer Liaison
4	Zhiya Zhang	540310485	zzha0323	Programmer
5	Zichen Lian	530816762	zlia0017	Doomsayer
6	Litong Shu	530588249	lshu6392	Programmer
7	Mengchen Liu	540223439	mliu0949	Tester
8	Tzu-Hsiang Tseng	540663646	ttse0054	Tracker

Attendances: Litong Shu, Shuhao Pei, Jiayue Wu, Haoyu Deng, Zhiya Zhang, Zichen Lian, Mengchen Liu,

Tzu-Hsiang Tseng

Apologies: None

Submission Date: 06/09/2026

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Weekly Group Meeting Minutes

Time: 05/09/2026

Venue: Teams

Meeting Minute Taker: Mengchen Liu

#	Agenda Item	Description/ Comments	Decision/Action	Who?	Items for escalation
1	What has been completed?	Core pages and basic navigation have been completed.	Review completed features.	Whole team	None
2	What is in progress?	Some page details and system connections are still being developed.	Continue implementation .		
3	What is working well?	Current functions are generally stable during basic testing.	Keep current structure.		
4	What needs improvement?	Some layouts and interactions still need refinement.	Improve usability and consistency.		
5	Reminders	Team members should update progress and report issues promptly.	Maintain regular communication.		
6	What needs to be completed before the next meeting?	Remaining functions should be completed before the next meeting.	Finish assigned tasks and test them.		

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7	What's Next	Integrate all modules and conduct overall system testing.	Prepare for next development stage.		
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Main Discussions

- The team discussed and allocated the development tasks for the coming week, mainly focusing on front-end and back-end development.
- Responsibilities were assigned to different team members to ensure that the development could progress efficiently.

Key Takeaways

- The main priority for the next week is to continue implementing the front-end and back-end functions.
- Team members should complete their assigned tasks and communicate with the team if any problems occur during development.
- Regular communication will help the team keep the project on schedule.

What Happened

- Tasks for the next stage of development were allocated among team members.
- The team reviewed the current development progress and discussed the work that still needs to be completed.
- Prepare next week's demo and presentation

What's Next

- **Jiayue Wu:** Further improve the tags search and navigation functions, refine page connections, and support system integration testing.
- **Shuhao Pei:** Continue refining the access-control system, integrate permissions with the text editor, and test different user roles.
- **Litong Shu:** Continue monitoring team progress, update the Status Checking Report, and coordinate remaining development tasks.

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- **Haoyu Deng:** Continue improving the login page and assist with authentication testing and team integration.
- **Zichen Lian:** Further test the Register Function, improve validation and error handling, and ensure stable backend integration.
- **Zhiya Zhang:** Continue improving the User Profile and Edit Profile functions and support backend and authentication integration.
- **Mengchen Liu:** Continue developing entity relationships, improve Knowledge Graph features, and perform further integration testing.
- **Tzu-Hsiang Tseng:** Continue improving the Home Page and develop the World Page based on client feedback.

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Weekly Client Meeting Minutes

Time: 05/09/2026

Venue: zoom meeting

Meeting Minute Taker: Jiayue Wu

Attendances: Litong Shu, Shuhao Pei, Jiayue Wu, Haoyu Deng, Zhiya Zhang, Zichen Lian, Mengchen Liu,

Tzu-Hsiang Tseng

Apologies: None

#	Agenda Item	Description/ Comments	Decision/Action	Who?	Items for escalation
1	What has been completed?	Home Page, tag search, User Profile Page, and entity page prototype completed.	Continue integration and testing.	The whole team	None
2	What is in progress?	Entity details, relationships, text content, and editing functions are still in progress.	Continue entity and relationship development.		
3	What is working well?	Tag search and the search bar work well and satisfied the client.	Extend search functions to worlds.		
4	What needs improvement?	World pages need separate layouts, and entities should support multiple relationships.	Create World Page and improve relationships.		
5	Reminders	AI assistant must be completed before semester end.	Track progress and report delays early.		
6	What needs to be completed	Complete text editing functions and the new World Page.	Prepare features for client review.		

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	before the next meeting?				
7	What's Next	Add editing, management, and contribution levels if time allows.	Start after core features are stable.		
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Main Discussions

- The team demonstrated the progress made during the week, including the Home Page, tag search, User Profile Page and the prototype entity page.
- The client provided feedback on the World Page design, suggesting that each world should have a separate repository-style page containing its description, rules and entities.
- The team and client also discussed how to represent relationships between entities, with support for multiple relationships.

Key Takeaways

- The current tag search and search functionality works well and meets the client's expectations.
- A separate World Page should be developed instead of directly opening the search page.
- Entities should be able to connect to multiple other entities through different relationships.
- The AI assistant is still an expected project feature before the end of the semester.

What Happened

- The team demonstrated the completed Home Page, search/tag filtering, User Profile Page and entity page prototype.
- The client reviewed the current system and gave positive feedback on the search functionality.
- The client suggested improvements to the World Page and entity relationship design.
- The team confirmed the main development priorities for the next week.

What's Next

- Develop the World Page, text editing, and entity relationship functions.
- If time allows, begin implementing editing and contribution permissions.

Individual Progress Report

Name: Jiayue Wu

Student ID: 530816072

Goals Complete the tags search and navigation functions, and improve the connection between related pages.

Progress Made Implemented the tag search and filtering functions. Basic navigation between related content is working correctly.

Blockers to Progress/Issues No major blockers. Some navigation details still need adjustment to match the overall system structure and client feedback.

Name: Shuhao Pei

Student ID: 530748773

Goals Define the system architecture and world-level access controls, and provide a Proposal API for text-editor integration.

Progress Made Implemented session authentication, four user roles, and public/private world visibility. Completed the proposal workflow for drafting, submission, review, and publication, with API documentation for teammates.

Blockers to Progress/Issues No major technical blockers. Text-editor integration requires coordination with the responsible teammate.

Name: Litong Shu

Student ID: 530588249

Goals: Review the current project progress, understand the implementation status of different system components, and complete the Status Checking Report.

Progress Made: Reviewed the current development progress and gained a clearer understanding of the features completed by different team members. Collected and organised the current project status and completed the Status Checking Report based on the team's progress.

Blockers to Progress/Issues: No major blockers. Further coordination with team members may be needed to keep the project status updated as development continues.

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Name: Haoyu Deng

Student ID: 540727454

Goals: help team do the collaborate job and improve the login page

Progress Made: improve the login and help group do the collaborate job.

Blockers to Progress/Issues: no many complex issues

Name: Zichen Lian

Student ID: 530816762

Goals: Implement and test the Register Function, including the registration page, form validation, backend API integration, and successful account creation.

Progress Made: Completed the Register page and connected it to the existing /register API. Added confirm-password validation, duplicate email handling, and password visibility controls. Tested successful registration, database insertion, and login using a newly created account.

Blockers to Progress/Issues: No major blockers. Some local setup issues occurred after the team updated the database schema and frontend dependencies, but these were resolved by installing the latest dependencies and importing the updated worldbuilding database schema.

Name: Zhiya Zhang

Student ID: 540310485

Goals Complete the development of the User Profile page and Edit Profile functionality, and continue supporting team communication and project documentation.

Progress Made Completed the User Profile page and Edit Profile function, including displaying user information and allowing users to update their profile details. Also assisted with client communication and reviewed team documentation to improve clarity and formatting. The main development goal for this week was achieved.

Blockers to Progress/Issues Further profile-related development and integration depend on the progress of the backend, database, and authentication functions. Coordination with other team members is required to ensure the frontend profile functions work correctly with the shared backend and user data structure.

Name: Mengchen Liu

Student ID: 540223439

Goals :Develop the Entity and Relationship system to allow users to search, view, and navigate between interconnected world entities. Establish a clear relationship structure that can support the project's Knowledge Graph functionality.

Progress Made: Implemented the entity detail pages and connected them with the backend API. Added support for displaying and navigating relationships between entities, including directional relationships such as **OWNS** and **BELONGS_TO**. Tested the system using multiple connected entities to ensure relationships and navigation work correctly.

Blockers to Progress/Issues : Integration between frontend components, backend APIs, and database relationship data also required additional testing. Further work is still needed for relationship creation/editing and more advanced Knowledge Graph features.

Name: Tzu-Hsiang, Tseng

Student ID: 540663646

Goals Develop the Home Page and ensure that the main page layout and navigation work correctly.

Progress Made Improved the Home Page layout and navigation and developed the repository-style World Page based on client feedback. Separated the World Page from the World Workspace and improved navigation between World Pages, Entity Pages and World Workspaces.

Blockers to Progress/Issues No major blockers. Some details of the World Page layout and content still need to be coordinated with the team.

1st UG CAPSTONE PROJECT STATUS CHECKING (Week 5)

Project Title:	Development of an AI-Assisted Collaborative Worldbuilding Platform for Game Design
Project Client:	Frank Lai
Group Tag:	COMP3888_M17_02_ZL

Project Information	Summary of report & feedback
Overall Information	
Brief on your project requirements, scope, and your proposed computational solutions based on Group's understanding	The project aims to develop an AI-assisted collaborative worldbuilding platform where users can create, organise, browse and expand fictional worlds. The main scope includes Wiki-style world and entity management, user accounts, collaborative contribution and access control, structured relationships between entities, search and navigation, and an AI assistant that uses existing world information to provide context-aware suggestions. The platform is being developed as a full-stack web application using Next.js, React and TypeScript. The backend manages users, worlds, entities, relationships, permissions and contributions, while the database stores the structured worldbuilding information. Entity relationships form the basis of the knowledge base, and the planned AI assistant will retrieve relevant world information to support consistent content generation. Human creators will retain control over accepting and managing contributed or AI-generated content. Client's feedback: No specific concerns or additional feedback were raised by the client
Project milestones and member's work splits and responsibilities	The current project milestone focuses on the development of the core platform functions and preparing these components for later integration. The main development tasks include the Home Page and navigation, user registration and profile management, tag-based search and navigation, entity pages and relationships, and access control. After these core components are developed, the project will move towards integrating the different modules, improving system connections and interactions, and conducting overall system testing. The development work is divided among team members based on these core functions. Tzu-Hsiang Tseng is responsible for the Home Page layout and navigation; Zichen Lian for user registration, form validation and backend API integration; Zhiya Zhang for the User Profile Page and profile management; Jiayue Wu for tag search and navigation; Mengchen Liu for the Entity Page, entity details and relationships; and Shuhao Pei for the Access Control system and role-based permissions. Other team members support the development by reviewing the overall system structure, testing the functions and providing suggestions for further improvement. Client's feedback: No specific concerns or additional feedback were raised by the client

Status Highlight	Summary of Project Progress
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Overall Project Status	
Progress and Achievements	The project has made significant progress in implementing the core platform functions. The team has completed the Home Page, search functionality, tag-based filtering, basic User Profile page and profile editing functions. Users can view worlds they have created or manage, explore existing content, and update profile information with password verification. A prototype of the detailed Entity Page has also been developed.
	The team has also started implementing relationships between entities, allowing users to navigate between connected world elements. Current development is moving toward adding text content and editing functions, improving entity relationships, and creating a dedicated World Page that contains the world's description, rules, entities and related information.
	Client's feedback: Frank was positive about the progress demonstrated and specifically stated that the search bar and tag-search functionality worked well and met his expectations. He suggested extending the search functionality so that users can also search across different worlds. Frank clarified that each world should have its own repository-style World Page, rather than directing users immediately to a search page. The World Page should contain information such as the world description, basic setting and rules, as well as its entities. He also confirmed that the current simple approach to entity relationships is sufficient, provided that one entity can connect to multiple other entities through different relationships. Frank agreed with the team's next-step plan to implement the text content/editor and the new World Page. He also reminded the team that the AI assistant remains an expected core feature and should be implemented before the end of the semester. Overall, he recognised that the team had made substantial progress and was satisfied with the direction of the current development
Key Issues	
Obstacles & Risks	At the current stage, there are no major blockers, but several integration and coordination risks remain. Some functions still depend on connections between the frontend, backend, database and authentication system, so changes to shared APIs or data structures may require additional coordination and testing. The entity relationship system also needs further development to support multiple relationships and relationship editing correctly. In addition, the team still needs to complete and integrate several core functions, including the World Page and text editing, before moving to overall system testing. The AI assistant has not yet been implemented and remains an important later-stage requirement, so development progress needs to be monitored to ensure sufficient time for AI integration and final testing before the end of the semester.
	Client's feedback: No specific concerns or additional feedback were raised by the client

Individual Engagement & Contribution (up to 0.5 page / member)

Shuhao Pei	
Participation and attendance of meetings	client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	Team Manager / Team Leader, System Architecture Designer. Coordinated team and client meetings, prepared agendas, and allocated tasks to keep development aligned with project requirements. My main technical contribution was designing and implementing the platform's access-control architecture and establishing

	the API contracts for collaborative content editing. Designed a world-level permission model with four roles—Owner, Manager, Author, and Reader—separating ownership, administration, content creation, and read-only access. Combined these roles with public/private world visibility, with entities and relationships inheriting their parent world’s permissions. Implemented server-side session authentication and permission checks across content retrieval, search, membership management, and editing operations. Developed the Proposal API to support private drafts, submission, withdrawal, review, and publication. Added revision checks, published-content version checks, and transactional approval to prevent conflicting edits and duplicate publication. Provided concise API documentation and a basic editing interface so teammates could develop the text editor against a shared contract.
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Zichen Lian	
Participation meetings	Participation of client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	Doomsayer, Full-stack Developer, Meeting Minutes Contributor My main contributions to the project have been in technical architecture, permission-related requirements, and the user registration function. During the proposal stage, I was responsible for the Methodologies and Feasibility / High-Level Technical Architecture section. I designed and presented the system architecture, including the frontend, backend, authentication service, database layer, Prisma ORM, and external Generative AI service, and explained how these components interact within the platform. As one of the Doomsayers, I also contributed to identifying potential risks, technical issues, and implementation concerns that could affect the project’s progress. During the implementation stage, I was responsible for the Register Function. I developed the registration page to match the existing login-page style, connected it to the backend /register API and MySQL database, and added confirm-password validation, duplicate-email error handling, and password visibility controls. I tested the complete registration workflow, including successful account creation, database insertion, bcrypt password storage, and successful login using a newly created account. I also contributed to meeting-minute preparation, Git/GitHub branch management, local development setup, database schema updates, and testing newly integrated team features after changes were merged into the main branch. My work helped connect the project’s technical design with a functional user-account feature and supported smoother integration between the frontend, backend, and database.

Zhiya Zhang	
Participation meetings	Participation of client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	Programmer / Testing and Analysis Contributor / Client Communication Support. Responsible for the Testing and Analysis section of the project proposal. Developed the project’s multi-level testing strategy, covering unit testing, integration testing, end-to-end testing, AI content validation, and regression testing. The testing plan was

	designed to verify core functions such as world and entity management, permissions, collaboration workflows, database operations, and AI-related features. Contributed to the web application development by designing and implementing the User Profile page and Edit Profile functionality. This included building the profile interface, supporting user information display and editing, and connecting the frontend profile functions with the backend logic required to update user information. Supported communication between the project team and the client by drafting and replying to emails, arranging meeting-related communication, and helping communicate project progress and requirement questions clearly. This helped maintain consistent communication with the client throughout the project. Also contributed to the quality of team deliverables by reviewing, editing, and formatting written submissions, improving language clarity, consistency, structure, and overall presentation before submission.
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Haoyu Deng	
Participation meetings	Participation of client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	- Developed the project introduction, aim, project question, scope, and timeline for the proposal. - Helped define the three core project functions: Wiki-style worldbuilding, GitHub-style collaboration, and the knowledge base with AI assistant. - Contributed to the initial SQL database and login/register backend design. - Helped translate client requirements into clear project objectives and development priorities.

Litong Shu	
Participation meetings	Participation of client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	Programmer / Tracker / Status Checking and Project Documentation Contributor. My main contribution during the proposal stage was working on the Data Collection and Computational Solutions section. I identified the main data sources for the platform and helped explain how this information could form the knowledge base of the system and support the AI-assisted worldbuilding functions. I also contributed to the high-level computational solution for the project by proposing a full-stack web application structure connecting the frontend, backend, database, authentication, and Generative AI services. This helped translate the project requirements into a clearer technical approach and supported the team's discussion of how different system components could work together.

	During the current development stage, I have been responsible for reviewing and checking the overall project status. I reviewed the progress of different team members, gained an understanding of the functions that have been implemented, and organised this information into the Status Checking Report. In addition, I participated in group and client meetings, followed changes in project requirements and client feedback, and communicated with team members to keep the project documentation consistent with the actual development progress.
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Tzu-Hsiang Tseng	
Participation meetings	Participation of client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	Tracker / Database and Frontend Contributor. Responsible for tracking team progress, task completion, and project development throughout the project. My main technical contribution was designing and implementing the worldbuilding database schema in MySQL, including the core structures for worlds, world members, entities, relationships, tags, and contributions. The database structure supports the project's knowledge graph, collaboration, and worldbuilding functions. I also contributed to the frontend development of the World Page based on client feedback. I implemented the repository-like World Page structure, allowing users to view world information and entities within a dedicated world page. I separated the World Page from the World Workspace so that general world content and collaboration/management functions have clearer purposes. In addition, I worked on navigation between World Pages, Entity Pages, search results, and World Workspaces, and tested these flows to ensure users could move between related content correctly. I also made interface and layout improvements to improve the clarity and consistency of the World Page.

Mengchen Liu	
Participation meetings	Participation of client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	Responsible for the development of the Entity and Relationship functions. My main contribution was implementing the entity search and detail pages, allowing users to view entity information and navigate between related entities. I also worked on the relationship structure between entities, including directional relationships such as OWNS and BELONGS_TO, and tested different relationship scenarios to ensure they were displayed correctly. This helped establish the basic Knowledge Graph structure of the project and made connections between world entities clearer and easier to navigate. In addition, I worked with other team members to integrate the entity functions with the existing frontend and backend, tested the functionality with different entity and relationship data, and participated in group and client meetings to discuss improvements and project progress.

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Jiayue Wu	
Participation meetings	Participation of client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	Responsible for the design and implementation of the Tags Search & Navigation functions. My main contribution was developing the tag-based search and filtering features, allowing users to quickly find relevant entities based on selected tags and keywords. I also worked on the navigation flow between the home page, world content, search results, and entity detail pages to ensure users could move between related content smoothly. During development, I tested the search and filtering functions with different inputs and adjusted the interface and navigation logic to improve usability and consistency. I also worked with other team members to integrate my functions with the existing front-end structure and ensure that the search and navigation components worked correctly with other completed pages. In addition, I participated in both group and client meetings, reviewed the overall system structure, provided feedback on interface design and functionality, and discussed improvements based on client feedback. I also assisted with general system testing and helped identify navigation and integration issues before further development.